1.Student Details

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About me:I am a student at IIT MADRAS BS program in diploma level.

2. Project Details

Project Title

Campus Placement Portal

Problem Statement

The goal of this project is to design and develop a web-based placement management system that connects students, companies, and administrators

on a single platform. The system allows companies to post job opportunities, students to apply for those jobs, and administrators to manage users and recruitment activities. The application simplifies the placement process by providing a centralized interface for job postings, applications, and candidate management.

Approach

The system is implemented as a web application using the Flask framework. It follows a modular architecture where different components such as user authentication, job management, and application tracking are handled through

separate routes and templates. Students can browse and apply for jobs posted by companies. Companies can create and manage job postings and review applicants. Administrators have the ability to approve users and manage placement activities. The backend uses a relational database to store user information, job listings, and application records.

3. AI / LLM Declaration

I used ChatGPT to assist with debugging Flask routes, improving code structure, generating sample HTML templates, and preparing documentation sections of this report. The AI assistance was

primarily used for guidance and suggestions. The final design, implementation, and integration of the project components were completed by me. The estimated extent of AI usage in this project is approximately 15–20%, mainly for coding assistance and documentation formatting.

4. Database Schema / ER Diagram

The system uses a relational database consisting of four primary tables: User, Job, Application, and PlacementDrive.

User Table

id (Primary Key)

name

email

password

role (admin, company, student)

is_approved

Job Table

id (Primary Key)

title

description

status

company_id (Foreign Key referencing User)

Application Table

id (Primary Key)

student_id (Foreign Key referencing User)

job_id (Foreign Key referencing Job)

status

PlacementDrive Table

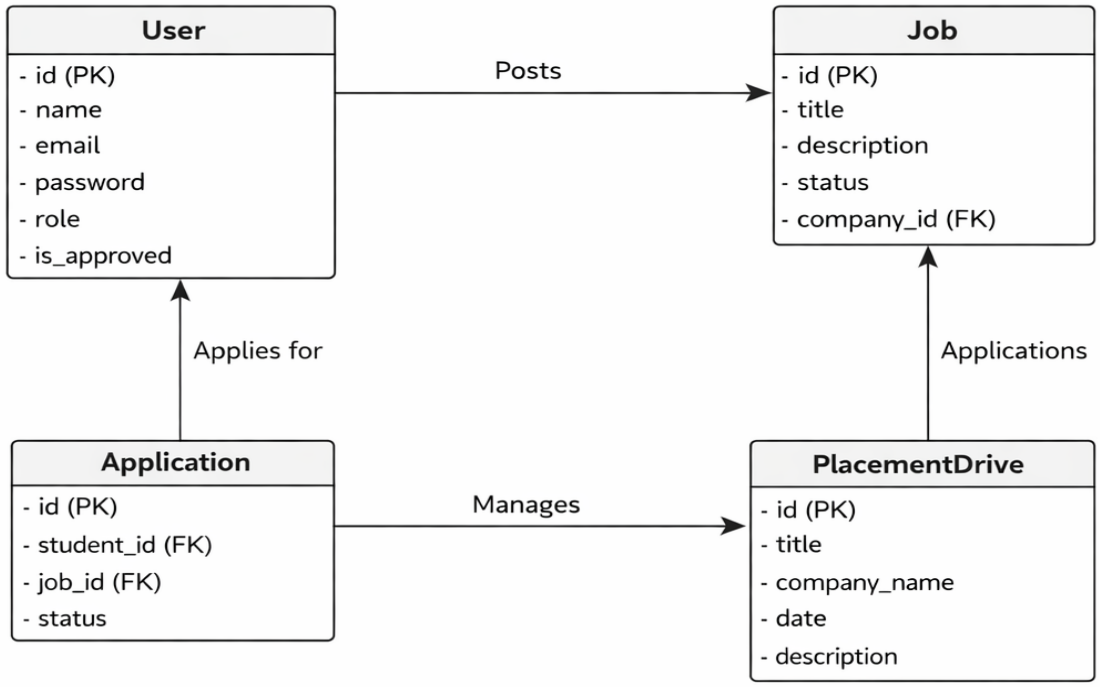
id (Primary Key)

title

company_name

date

description



5. API Resources / Endpoints

/register

POST

Registers a new user

/login

POST

Authenticates a user

/logout

GET

Logs out the current user

/admin/dashboard

GET

Admin dashboard

/company/dashboard

GET

Company dashboard

/student/dashboard

GET

Student dashboard

/company/add-job

POST

Company adds a new job

/apply/<job_id>

POST

Student applies for job

/admin/approve/<user_id>

GET

Admin approves a user

/admin/delete/<user_id>

GET

Admin deletes a user

/company/applicants/<job_id>

GET

Company views applicants

6. Technologies and Frameworks Used

Write something like this:

Programming Language

Python – Used for backend logic and application development.

Web Framework

Flask – Lightweight Python web framework used to build the application and manage routes.

Database

SQLite – Relational database used to store users, jobs, and application data.

ORM

SQLAlchemy – Used to interact with the database using Python models.

Frontend

HTML – Used for creating the structure of web pages.

CSS / Bootstrap – Used for styling and responsive design.

Version Control

Git & GitHub – Used for source code management and version tracking.